Biochemistry of Birds and Reptiles

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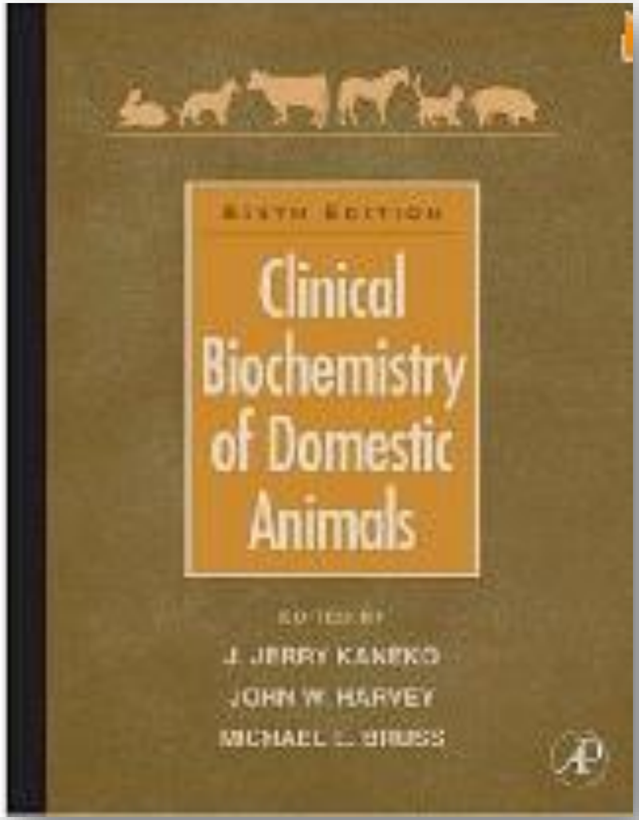
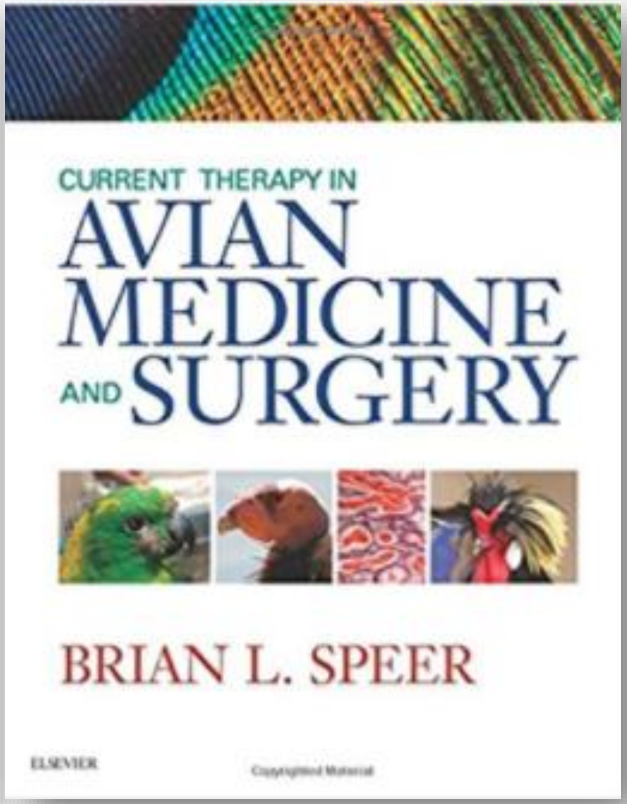
UCDAVIS
VETERINARY MEDICINE

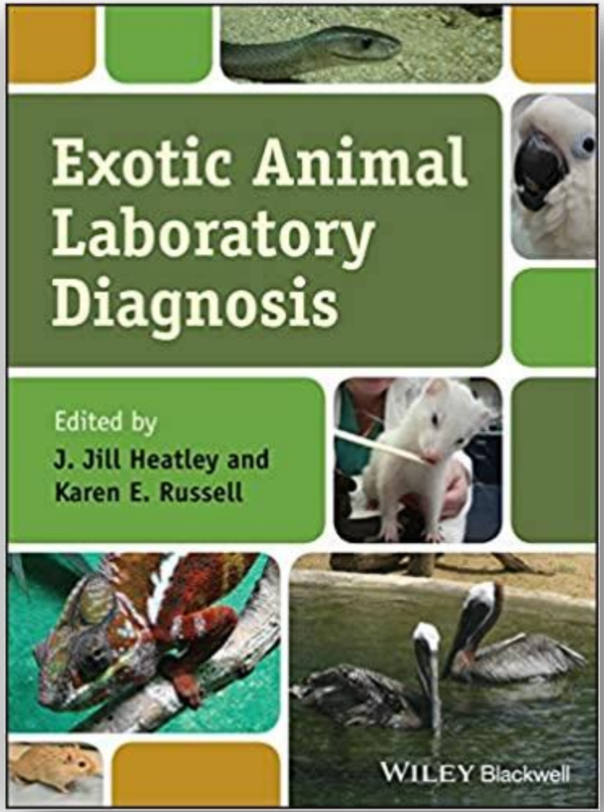
Introduction

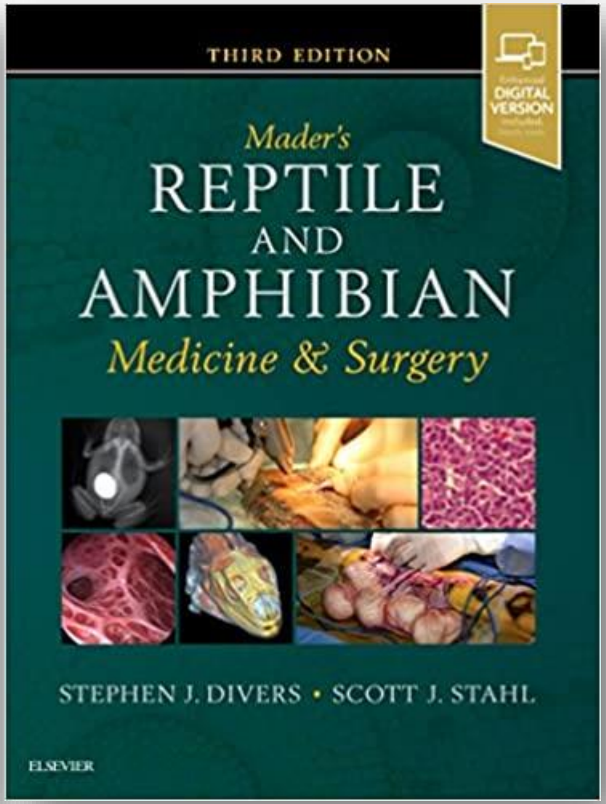
- Biochemistry is similar between birds and reptiles
- Compared to mammals:
  - Less sensitive & specific
  - More variable
  - More exceptions, especially in reptiles

Resources - Biochemistry







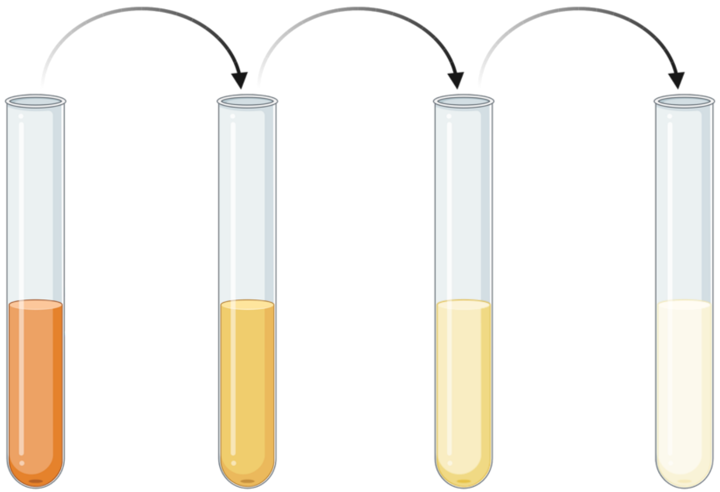


Sample Collection

- ASAP, before physical examination, before any injections or prolonged restraint
- Collect in red top if enough volume, otherwise, in heparin
- Centrifuge and separate plasma/serum quickly

Sample Collection

- If not enough volume, one can consider diluting the sample before running the biochemistry profile, but this is **not ideal**
- Dilution will affect most analytes except AST, CK, glucose, and uric acid



The Biochemistry Panel

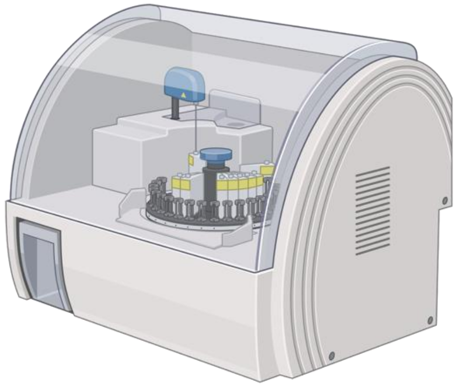
- Labs or in-clinic analyzers may offer biochemistry panels which are:
  - Incomplete
  - Contain parameters which are not useful in birds/reptiles
  - Same panel for all birds and reptiles (+/- all non-mammals)
- Low sample volume may limit the number of parameters you can run
  - You may be able to tell the lab the order of priorities

UCDAVIS VETERINARY MEDICINE

Proteins

Total Protein:

- **Biuret method:** A substance changes colour when it binds proteins → color change is measured/quantified.
- Used by most reference and in-clinic biochemistry analyzers
- This is what you get on biochemistry panels

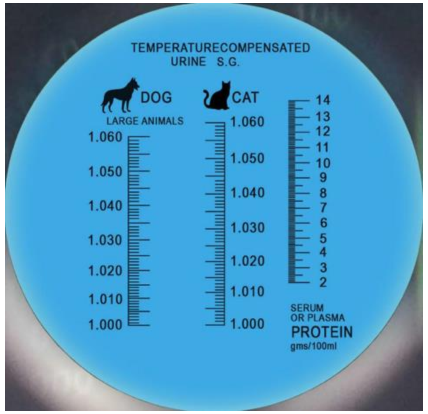


Biochemistry analyzer

Total Solids:

- Measured with a **refractometer**
- **“Estimate”** of total proteins
- Glucose, urea, cholesterol and triglycerides also contribute to this → to a greater extent in non-mammals than in mammals
  - Therefore, **not a good estimate** of total protein in birds/reptiles
- Usually reported as part of the CBC

Refractometer



Proteins

Albumin:

- **Bromocrescol green (BCG) method:** “Standard” method for measurement of albumin in people and common mammal species. Similar to biuret but more specific to albumin.
- **Not reliable** in non-mammals, and some exotic mammals!
- Albumin is usually underestimated (but depends on species)

Globulin:

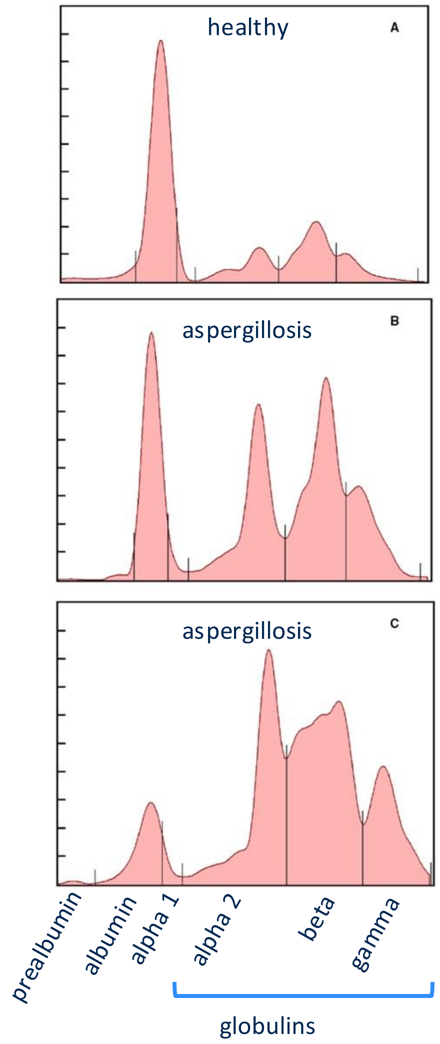
- Globulin is **not** directly measured, they are calculated:
  - **Total protein (biuret) – Albumin (BCG) = Globulin**
  - If albumin is not accurate, globulin is also not accurate

Proteins

Protein electrophoresis:

- Plasma or serum applied to a gel and separated by an electric current
- Peaks on electrophoretogram correspond to different groups of proteins including:
    - Prealbumin (some species only); albumin; and alpha, beta and gamma globulins

Agarose gel electrophoresis (AGE) electrophoretograms of heparinized plasma obtained from African penguin

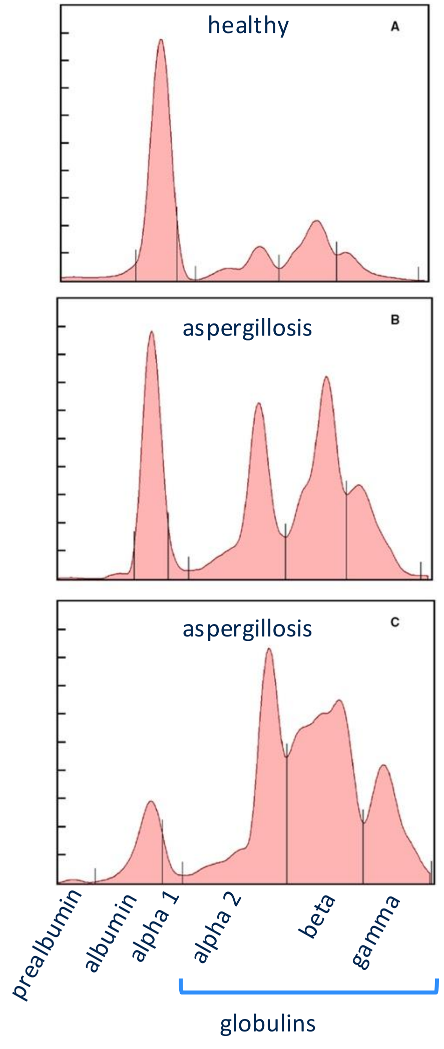


Proteins

Protein electrophoresis:

- Could be more accurate way to differentiate between albumin and globulins BUT some subjectivity to interpretation (eg: some fractions blend together)
- More research needed to document which proteins are found in each peak and therefore how to interpret changes
- Currently mostly used to detect evidence of inflammation

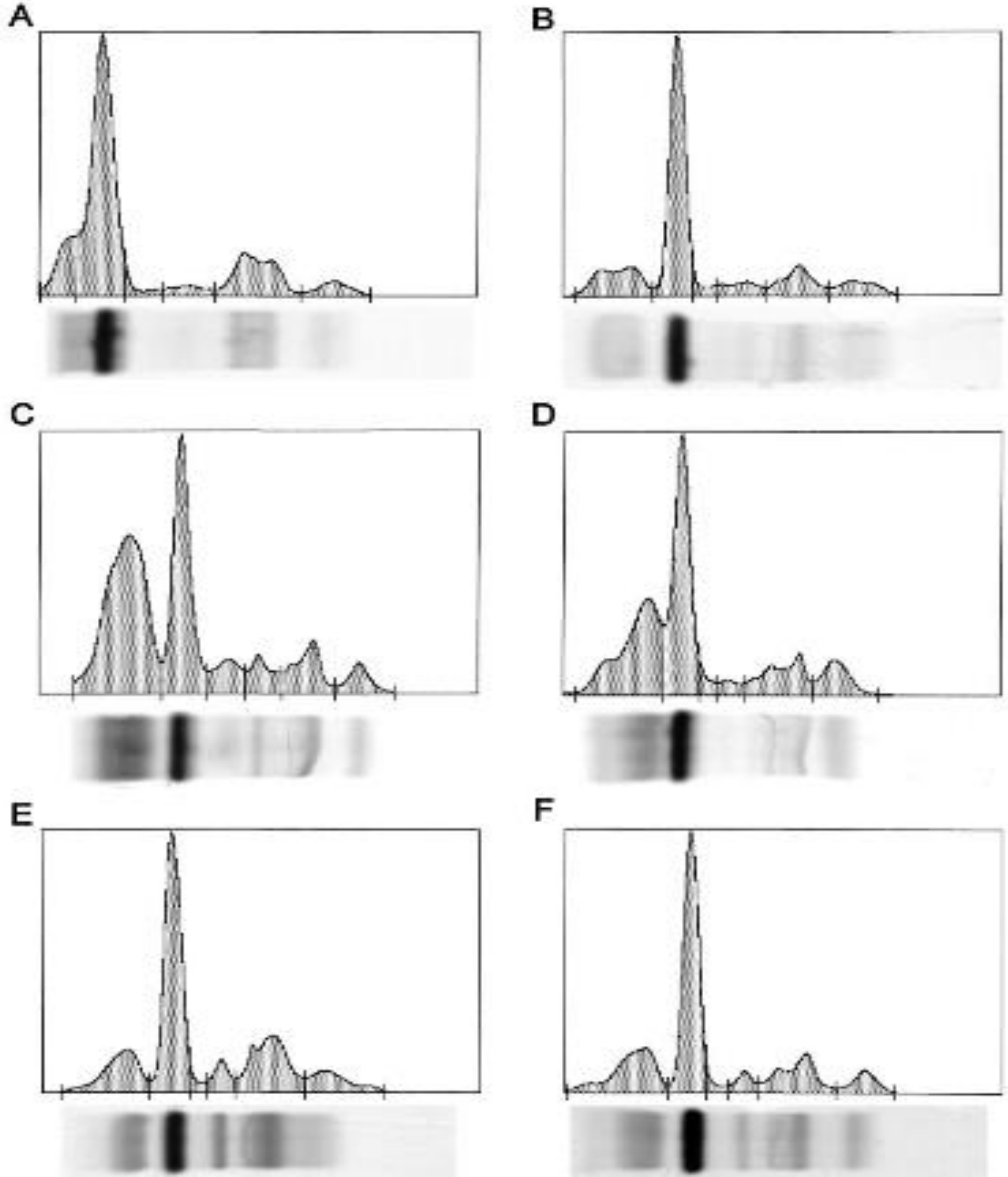
Agarose gel electrophoresis (AGE) electrophoretograms of heparinized plasma obtained from African penguin



Normal eletrophoretograms in various birds → interpretation need to be species-specific (if possible)

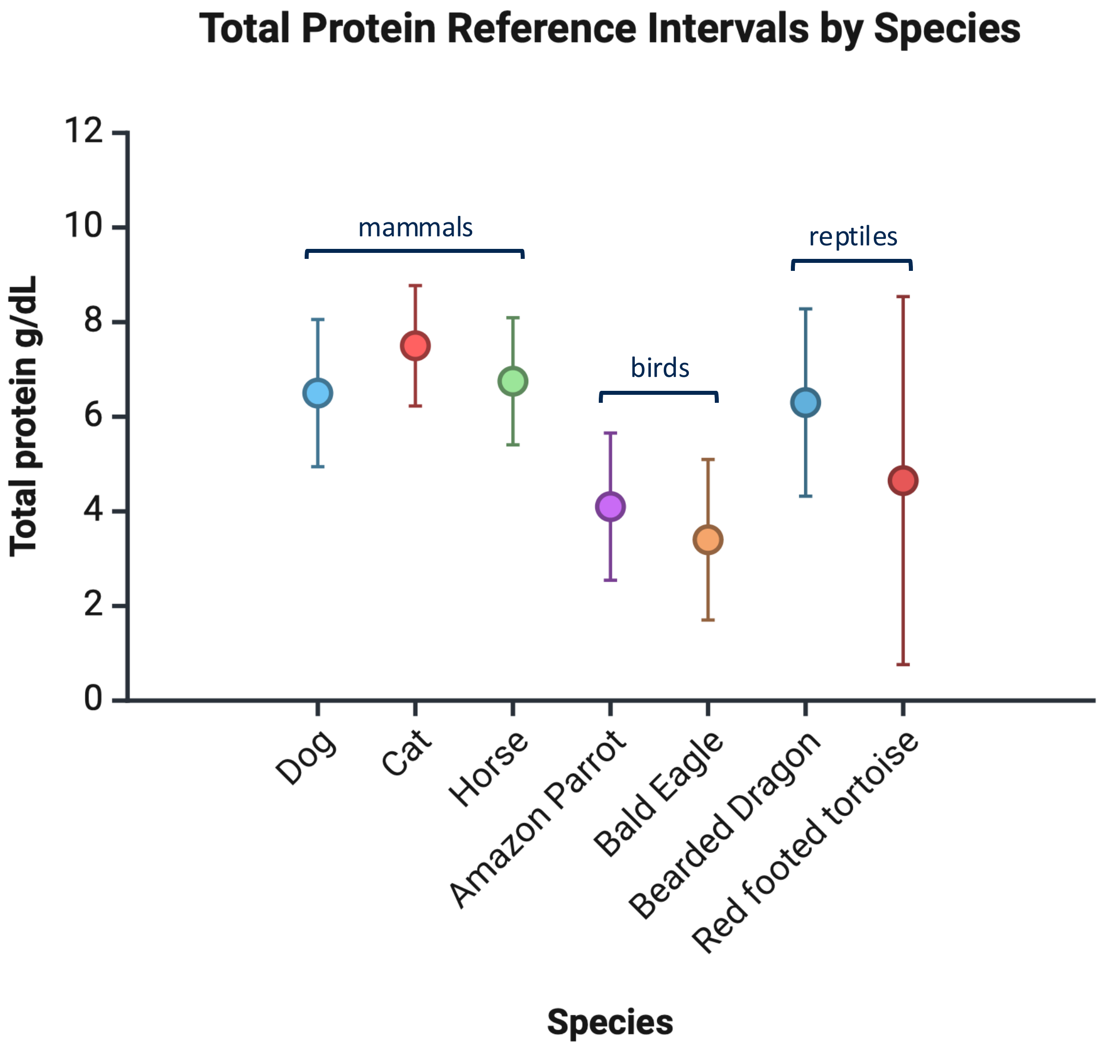
(A) African Grey parrot
(B) Amazon parrot,
(C) Budgerigar,
(D) Cockatiel,
(E) Cockatoo,
(F) Conure

Cray et al, 2007



Proteins

- Birds have lower total proteins than mammals
- Reptiles close to mammals or have wide ranges



Proteins

Increased TP	Dehydration Hemolysis Vitellogenesis Inflammation (increased globulins as positive acute phase proteins)
Decreased TP	Chronic disease (decreased albumin as negative acute phase protein) Emaciation Chronic protein loss Bleeding GI parasitism Molting (dilution effect)

Acute-Phase Proteins (APP)

Fibrinogen:

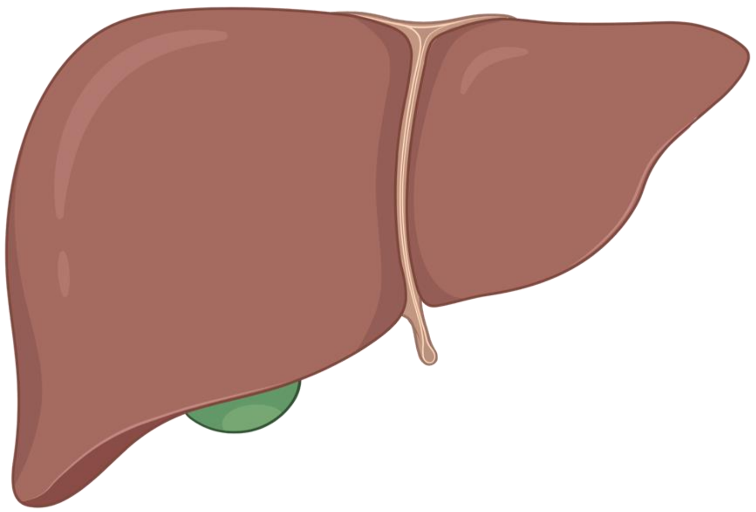
- Crude quantification by heat precipitation
- Total solids measured in blood before and after heat treatment

Others:

- Seldom used and few studies correlating with clinical diseases
- Quantified using immunoassays (aka: using antibodies) so usually species-specific
- Protein C-reactive not a major APP
- Serum amyloid A is a major APP

Clinical Enzymology

- Tissue enzyme activities determined in a limited # of species
  - Ideally should be done in all species
  - Extrapolating current knowledge to all species
- **Liver**
  - Enzymes used to detect liver **damage** and **cholestasis**
  - Several analytes needed to evaluate liver disease because:
    - None is highly specific and sensitive
    - They have different half-lives
    - They give different information



Green = useful; red = not useful

Simplified Tissue Distribution of Commonly Used Enzymes in Avian Biochemistry[19,38,39]

	Hepatic Tissue			Muscular Tissue			Other Tissues
	Detection	Sensitivity	Specificity	Detection	Sensitivity	Specificity	Detection
AST	++	+++	–	++	+++	–	-
GLDH	+++	+	+++	+	–	–	+++ (kidney but excreted in urine)
GGT	+	+	+++	–	–	–	+++ (kidney)
LDH	++	+	–	++	+	–	+
ALT	++	++	–	++	+++	–	+
ALP	+	–	–	++	–	–	+ (duodenum, bone, kidney)
CK	–	–	–	+++	+++	+++	-
NAG	+	–	–	–	–	–	+++ (6 times higher in kidneys)

Grosset, Beaufrere, clinical biochemistry, in: Speer B, Current Therapy in Avian Medicine, Elsevier, 2017

Enzyme	Half-Life
AST	7-8h
LDH	0.5-1h
GLDH	0.5-1h
CK	3-4h

Other notes:

- GGT: Induced expression during cholestasis, not useful in reptiles
- GLDH: Specific but not sensitive
- CK: muscle
- LDH: High concentration in RBCs

Conclusion: you need a **panel** of enzymes to get a full picture!

Clinical Enzymology

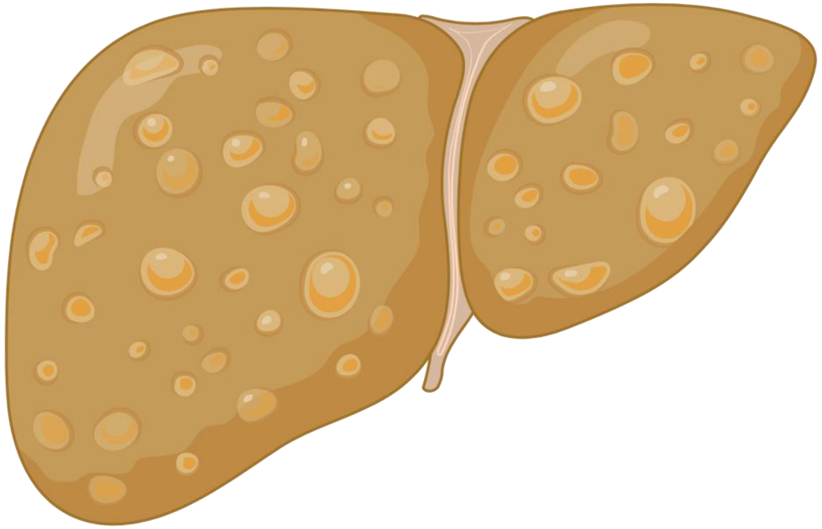
- Need to increase more than 2-3-fold to be considered clinically significant
- Previous **IM injection** may increase **AST** and **CK** for up to 3-5 days
- **Hemolysis** will increase **LDH**
- Minimal interference from lipemia (or will be flagged)

Clinical Enzymology

Enzymes have a **low sensitivity** for diagnosis of **chronic metabolic disorders** affecting liver, especially in reptiles.

Including:

- Hepatic lipidosis
- Iron storage disease
- Amyloidosis

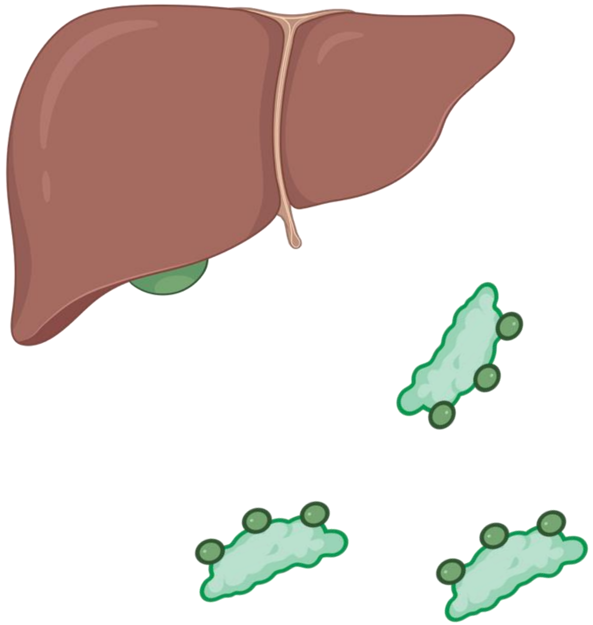


Hepatic Function

Bile acids: <u>high sensitivity and specificity!</u>

Elevations seen with:

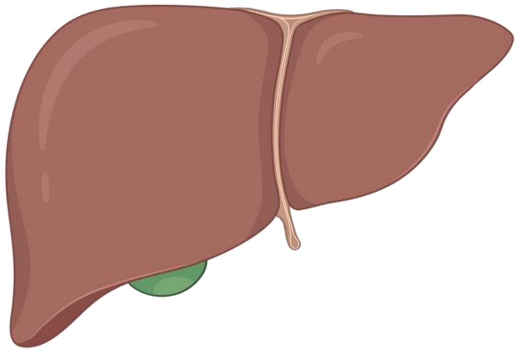
- Post-prandial
- Decreased liver function/liver failure
- Congestive heart failure



Hepatic Function

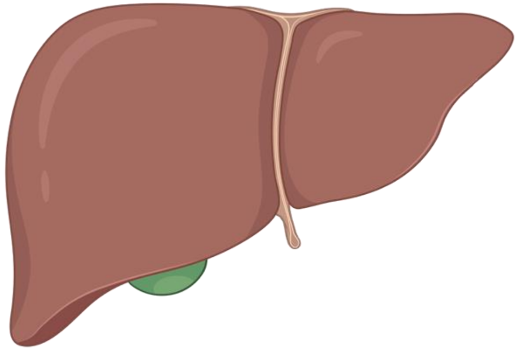
Bilirubin:

- In mammals, heme is broken down (after iron removal) into **biliverdin**, which is then converted to unconjugated bilirubin.
- In birds and reptiles, the last step does not occur (lack of biliverdin reductase) therefore bilirubin measurement is of **limited usefulness**.
- Bilirubin can occasionally be increased with severe liver disease



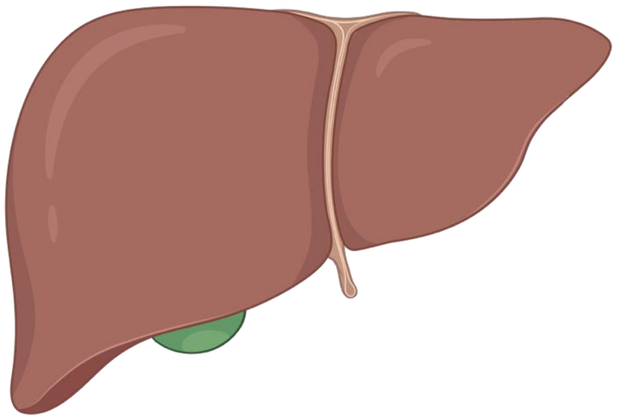
Hepatic Function

- **Biliverdin is not measurable**
  - Excreted quickly through the kidneys
  - Uncommon to grossly see hyperbiliverdinemia or green tinge to mucous membranes with liver disease/hemolysis
- Yellow plasma should not be interpreted as icterus
  - **Yellow color is usually food-related** (eg: carotenoids, yolk sac if eating chicks)
- Ammonia: usefulness unknown

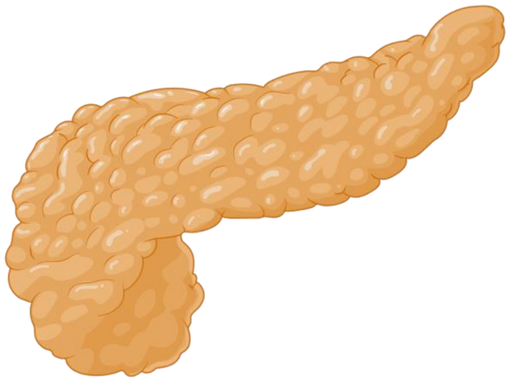


Hepatic Function

- Decrease in hepatic metabolites **uncommon, or with very advanced disease** (e.g. iron storage disease in mynahs).
- If present, decreased:
  - Uric acid
  - Glucose
  - Total protein
  - Cholesterol
- Increase in cholesterol with cholestasis



Pancreas

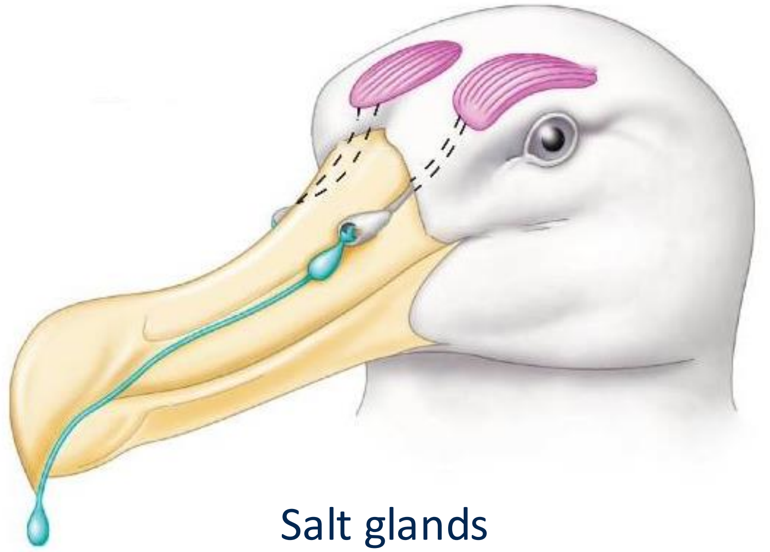


Amylase and lipase

- Not considered useful
- Seems to increase in **some (few)** pancreatitis and enteritis cases
- Pancreatitis uncommon
- Associations with pancreatitis:
  - Species: Quaker parrots
  - Zinc toxicosis
  - Avian chlamydiosis
  - Paramyxovirus 3 (PMV3) infections

Osmoregulation

- Kidneys
- Cloaca (coprodeum), colon, ceca
- Salt glands (salt excretion in sea birds)
- Bladder (especially chelonians, absent in snakes)

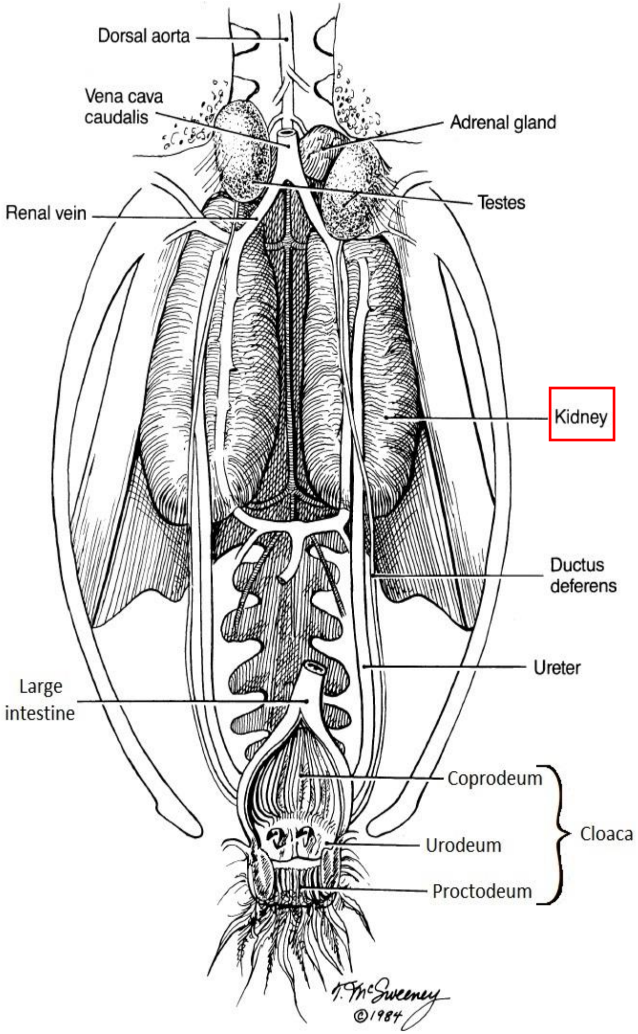


Salt glands

Osmoregulation

Kidneys:

- **Birds:** have 30% looped nephrons (looped = can concentrate urine) and rest are loopless
- **Reptiles:** all loopless nephrons
- Limited renal concentrating abilities
- **Post-renal handling of urine:** water can be resorbed from urine in urinary bladder, cloaca and colon

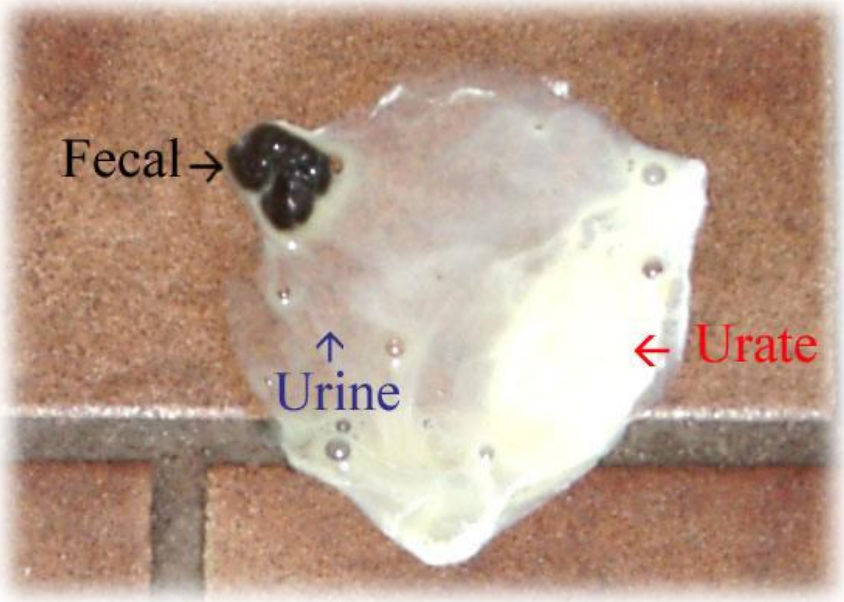


Bird kidneys

Nitrogen Metabolism

Nitrogen excretion:

- Predominantly in the form of **water-insoluble uric acid**
- **Water-soluble** urea is a smaller component (except in some species, esp. aquatic chelonians)



Droppings composed of feces, urine (clear, watery), and urate (white)

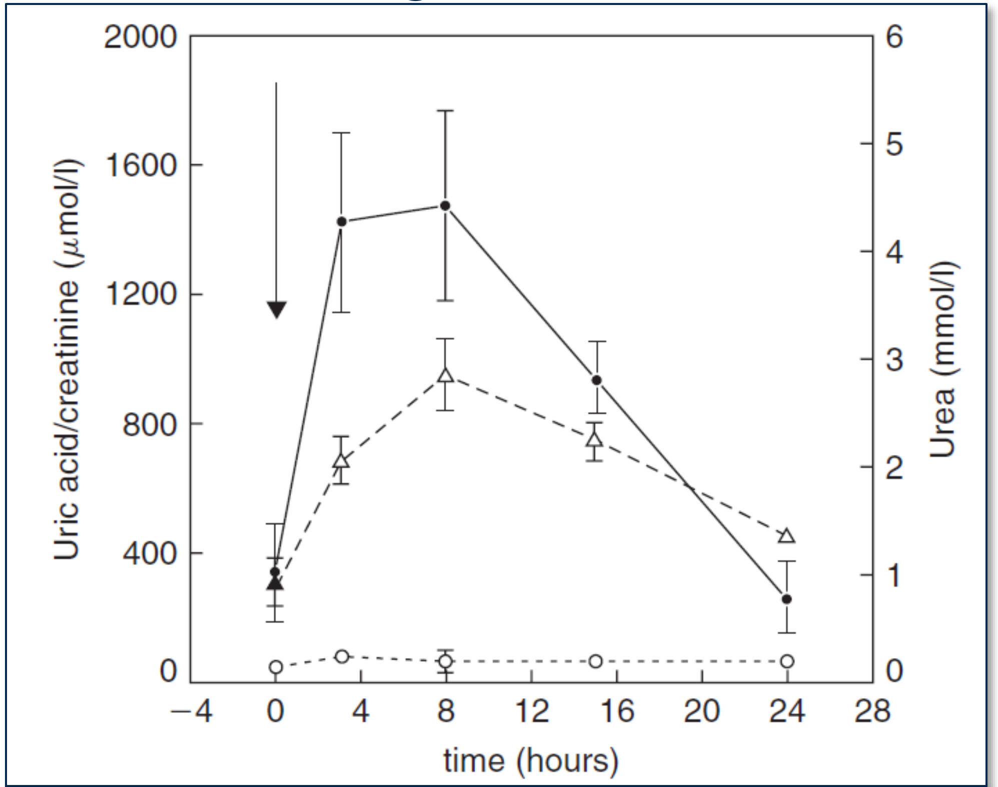
Nitrogen Metabolism

Uric Acid Increase (hyperuricemia)

- **Renal failure**: (need 70% nephron loss)
- Dehydration:
  - Active excretion of uric acid can continue with mild dehydration
  - Hyperuricemia happens with **moderate or severe dehydration**
- **Post-prandial** after **protein-rich meal**
- Uric acid can start to deposit in joints (gout) or in organs (eg: kidneys) and cause tissue damage

Renal Paramaters

Peregrine falcon



●, [uric acid] (μmol/L); Δ, [urea] (mmol/L);

Lumeij, in Kaneko, Clinical Biochemistry of Domestic Animals, Academic Press

Post-prandial increase in urea and uric acid:

- Seen in carnivorous, piscivorous, insectivorous species
- May be as high as pathologic increase
- May last more than 24h after a meal

Renal Parameters

Urea

- Increases with dehydration in birds (prerenal azotemia)
- Aquatic chelonians, tuatara and possibly desert terrestrial chelonians excrete significant urea:
  - Baseline urea is higher than in other species
  - May increase with renal failure

Red-eared slider: fresh-water turtle

Tuatara: a lizard from New Zealand

Renal Parameters

	Percent of total urinary nitrogen appearing as: Ammonia	Urea	Urates
Testudines			
Wholly aquatic	20–25	20–25	5
Semi-aquatic	6–15	40–60	5
Wholly terrestrial			
Moist environment	6	30	7
Dry environment	5	10–20	50–60
Gopherus agassizii (Desert tortoise)	3–8	15–50	20–50
Pseudemys scripta (Freshwater turtle)	4–44	45–95	1–24
Crocodilia	25	0–5	70
Squamata Sauria	insignificant—? highly significant	0–8	90
Ophidia	insignificant—? highly significant	0–2	98
Rhyncocephalia			
Sphenodon punctatus	3–4	10–28	65–80

Gans C, Biology of the reptilia, Vol. 5, 1976

Renal Parameters

Canthaxanthines:

- Naturally occurring carotenoid, similar to pigment in carrots
- In zoos, pigment is added to the food of some birds to make their colour more red/pink
- Seen mostly in these species
  - Flamingoes
  - Red factor canaries
  - Red ibises
  - Spoonbills
  - Etc…
- When present in plasma/serum you get invalid uric acid measurement on the Vetscan

Renal Parameters

Calcium-to-phosphorus ratio

- **Most sensitive parameter of renal disease in reptiles when inversed or decreased!**
- Not proven to be useful in birds
- Should be around 2:1
- Renal secondary hyperparathyroidism in reptiles

Renal Parameters

Dipstick

- Low clinical utility and few studies
- Dipstick only useful for glucose
- Birds’ main ketone is BHBA, not aceto-acetate, so dipstick won’t work
- Bilirubin pad not useful
- Other analytes of low value because of post-renal modification of urine and mixing with feces

Renal Parameters

- **Urine colour**
  - Green: biliverdinuria
  - Red: lead poisoning (porphyrin or hemoglobin) or capture myopathy
- **Urinary casts associated with renal diseases**

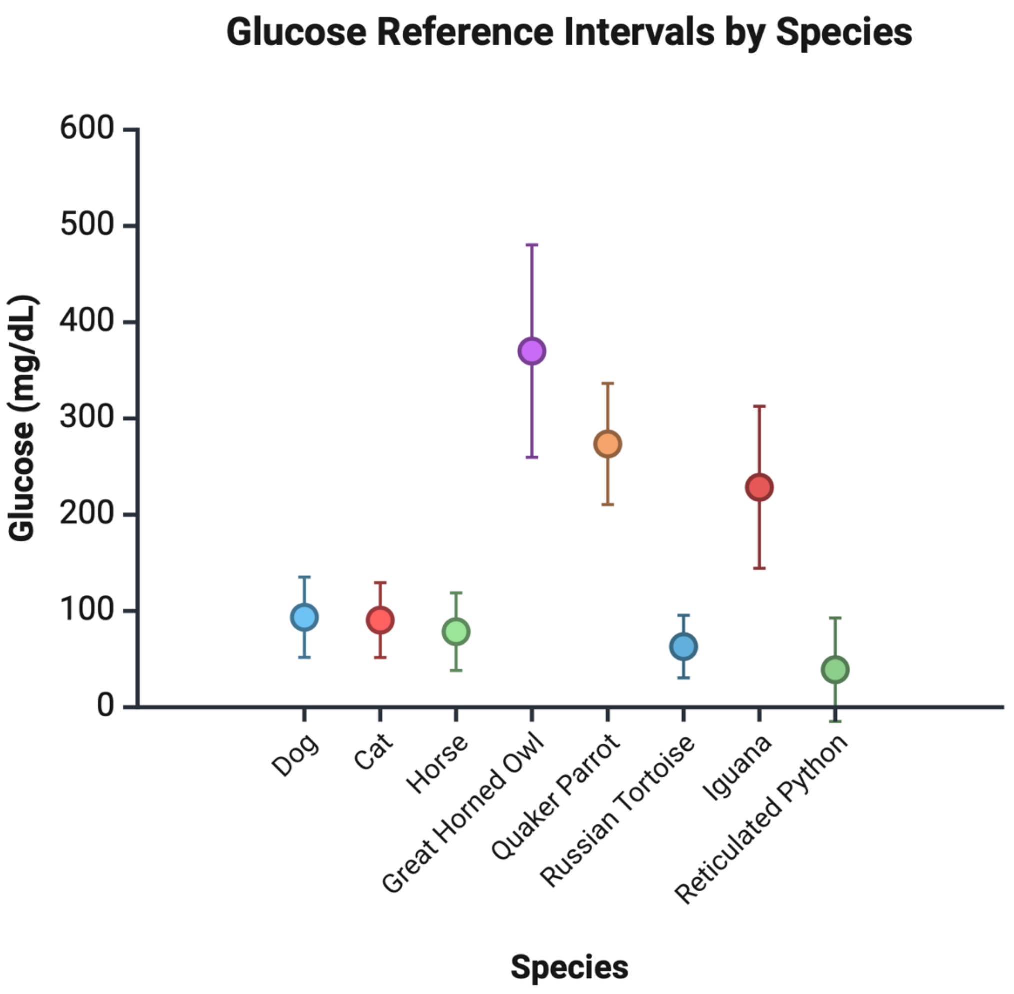
Glucose Metabolism

Avian blood glucose: ~ twice that of mammals (180 – 360 mg/dL)

- Birds have ~10 times less insulin and ~8-10 more glucagon than mammals

Reptile blood glucose: half that of mammals (54 – 90 mg/dL)

- But more variation in reference intervals across species in literature (eg: bearded dragons up to 300, chameleons up to 400 – Not fasted?)



Glucose Metabolism

- RBCs don’t depend on glycolysis for energy (fatty acids used)
  - Delayed separation of plasma/serum does not lead to spurious hypoglycemia as much as in mammals
- **Glucometers not reliable in birds**

Glucose Metabolism

Diabetes mellitus

- Common in mynahs and toucans
- Very high glucose (720-1080 mg/dL)
- Glucosuria (dipstick)
- Ketonuria not detected on dipstick and ketone-meters as main ketone is different
- Can measure BHBA and fructosamine on biochemistry analyzers

Mynah

Glucose Metabolism

- Differentials for causes of diabetes mellitus (birds):
  - Pancreatitis
  - Iron storage disease
  - Idiopathic type I
  - Zinc toxicosis
- Note that **stress hyperglycemia is common**

Glucose Metabolism

- **Differentials for hypoglycemia (birds):**
  - Anorexia and starvation in young bird/neonates (rare in adults)
  - Sepsis

Glucose Metabolism

- **Differentials for hyperglycemia (reptiles):**
  - Pancreatitis
  - Transient idiopathic (days to weeks)
  - Neuroendocrine gastric tumours in bearded dragons
- **Differentials for hypoglycemia (reptiles):**
  - Starvation
  - Improper husbandry (especially temperature)
  - Sepsis

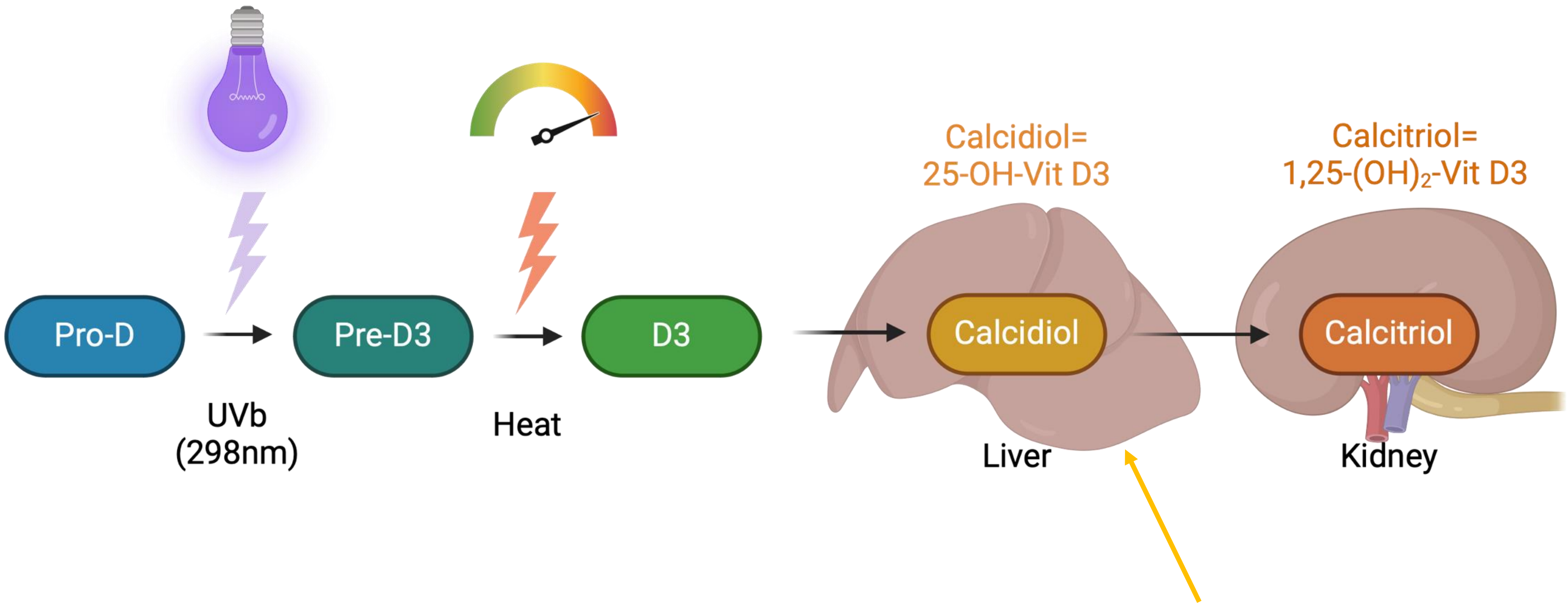
Lipid Metabolism

- **Dyslipidemia** frequent in parrots:
  - Quaker parrots are highly predisposed
  - Associated with various diseases:
    - Female reproductive diseases
    - Atherosclerosis
    - Nutritional imbalances
    - Hepatic lipidosis
    - Obesity, fatty tumors
- Lipids also increase **post-prandially** and with **cholestasis**

Calcium

- Similar regulation as mammals
- Ionized Calcium (iCa) in similar concentrations
- Female reproductive physiology
  - Hypercalcemia in reproductively active females (total Ca only, ionized Ca is stable)
  - High calcium exchange in the egg-shell gland (highest ion transport in animals)
  - Medullary bone serves as storage of calcium for egg production

Calcium: Reptile Vitamin D3 metabolism



- Secondary hyperparathyroidism associated with low calcidiol
- Calcidiol can be measured (decreased)
- PTH assay not reliable across species

Electrolytes: Calcium

Causes of metabolic bone disease in reptiles:

- Lack of UVb exposure
- Inversed Ca:P in diet
- Kidney disease
- Liver disease
- Low environmental temperature

Common conditions in birds and reptiles

Hypocalcemia	Hypercalcemia
• African grey parrots (idiopathic) • Zinc toxicosis • Nutritional hyperparathyroidism • Renal hyperparathyroidism (chameleons) • Sepsis (ionized) • Egg laying in deficient birds • Overheparinization (iCa) • Clot in sample	• Vitamin D toxicosis (macaws, reptiles) • Rodenticides • Dehydration • Reproductive physiology/diseases (iCa normal) • Lymphoma (very rare) • Lipemia

Electrolytes: Na, Cl, K

- Similar interpretations and differentials than in mammals
  - Except for salt gland adenitis in sea birds:
    → Increased ions because of impaired salt excretion
- Less tightly controlled in reptiles
- Hyperkalemia with hemolysis

Questions?

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